Meridian Pictures: Strategy and Slate Planning

ETC2420/5242 | Group 025 | Assignment 2

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1 Task 1: Slate Health Check

1.1 (a) Financial and Slate Summary

The course-provided dataset is based on The Movie Database. Films released after 2016 are excluded, while budgets below $10,000 and zero revenues are treated as missing. ROI is recalculated as revenue divided by production budget. For genre comparisons, films are classified using their first-listed genre to prevent double-counting.

```
focus_genres <- c("Action", "Comedy", "Drama", "Horror", "Animation")
movies <- read_csv("data/movies.csv", show_col_types = FALSE)

# Placeholder budgets (< $10,000) and revenues (<= 0) become missing; ROI is recomputed
movies_clean <- movies |>
  filter(!is.na(release_year), release_year <= 2016) |>
  mutate(
    budget = if_else(budget < 10000, NA_real_, as.numeric(budget)),
    revenue = if_else(revenue <= 0, NA_real_, as.numeric(revenue)),
    roi = if_else(!is.na(budget) & !is.na(revenue), revenue / budget, NA_real_),
    budget_m = budget / 1e6,
    franchise_status = factor(franchise_status, levels = c("Original", "Franchise"))
  )

movies_focus <- movies_clean |> filter(primary_genre %in% focus_genres)
franchise_summary <- movies_clean |>
  count(franchise_status, name = "Films") |>
  mutate(Percentage = round(100 * Films / sum(Films), 1))
```

Table 1: Data quality summary

Measure	Count
Films in original dataset	4803
Films in analysis period	4801
Missing or invalid budgets	1070
Missing or invalid revenues	1425
Missing ROI values	1587
Films with valid budget and ROI	3214

Table 2: Summary statistics for production budgets and ROI

Variable	N	Mean	Median	SD	Q1	Q3
Production budget ($m)	3731	37.39	24.0	42.69	8.85	50.00
ROI (multiple)	3214	11.09	2.3	239.95	1.03	4.42

Table 3: Franchise and Original film composition

franchise_status	Films	Percentage
Original	4541	94.6
Franchise	260	5.4

Of 4,801 films in the analysis period, 1,070 have an invalid or missing budget, 1,425 have missing revenue and 1,587 have missing ROI. Financial results therefore reflect only films with usable values. The median budget is **$24 million** and median ROI is **2.30**; medians are more informative than means because a few films have unusually large values.

1.2 (b) Distributions and Genre Comparisons

Budgets are right-skewed, with most films concentrated at the lower end. ROI is also highly skewed, so it is displayed on a log scale; the dashed line marks revenue equal to budget (ROI = 1). **Original films make up 94.6%** of the dataset, compared with **5.4% franchises**.

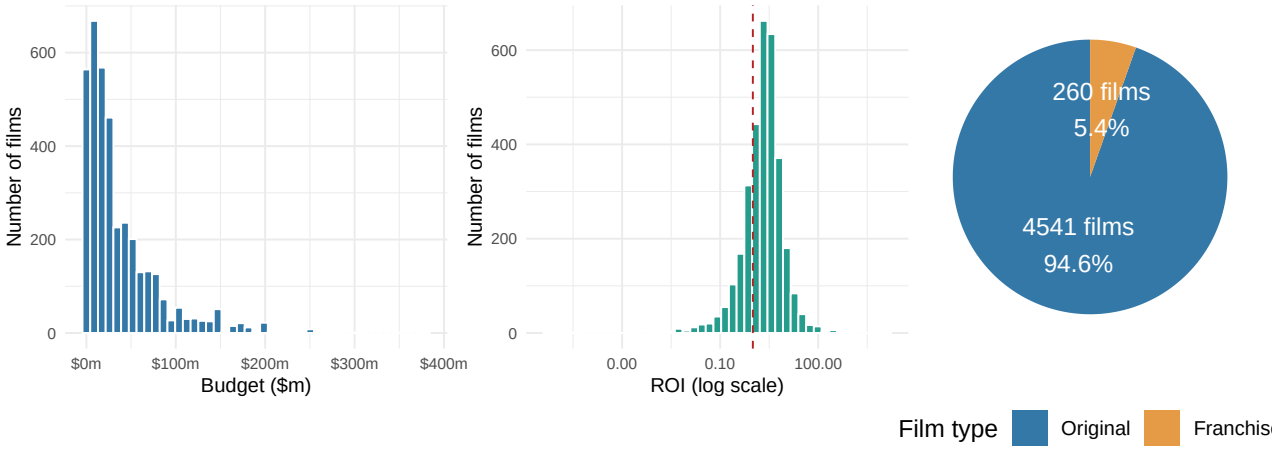


Figure 1: Distribution of budgets and ROI, and film composition.

1.2.1 Comparisons Across Genres

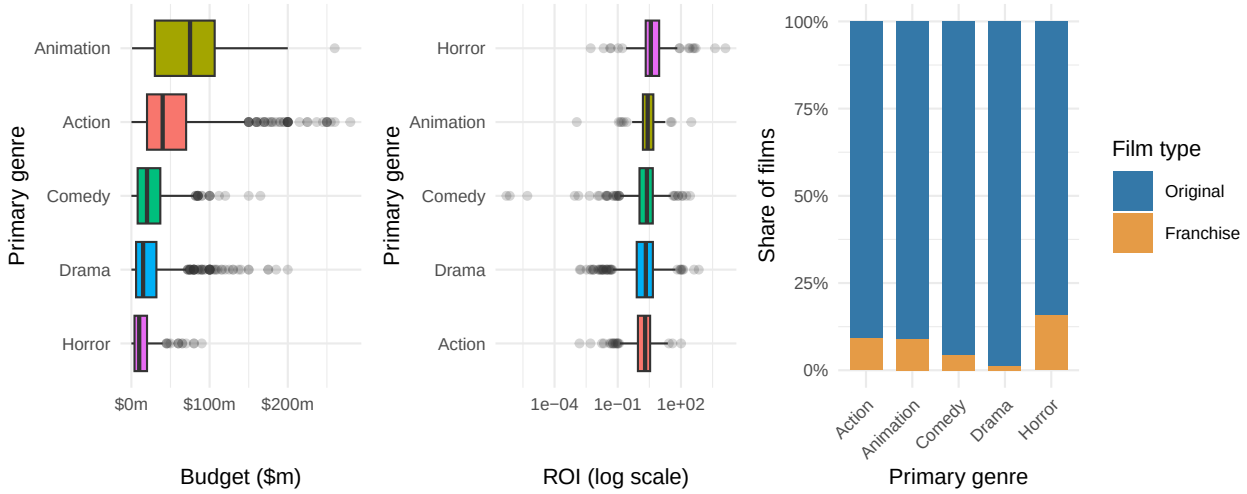


Figure 2: Budgets, ROI and franchise composition across the five focus genres.

Animation and action generally have higher production budgets, while horror and drama tend to be lower-budget. ROI overlaps substantially across genres, suggesting that a larger budget does not guarantee a greater revenue-to-budget multiple. Franchise shares also vary by genre, although original films dominate all five groups.

1.3 (c) Changes Over Time

Between 1990 and 2016, median production budgets fluctuate without a consistent upward trend. Median ROI also varies considerably across release years but generally remains above one, indicating that reported box-office revenues typically exceed production budgets. Original productions dominate the slate throughout the period, although the proportion of franchises changes between years. These patterns suggest that production costs and returns vary over time, without establishing a consistent shift towards franchises.

1.4 (d) Head of Production Summary

The dataset is dominated by original films, with relatively few franchises. Typical production budgets are well below the largest blockbuster budgets, while ROI varies considerably between films. This makes the median more useful than the mean for describing a typical project.

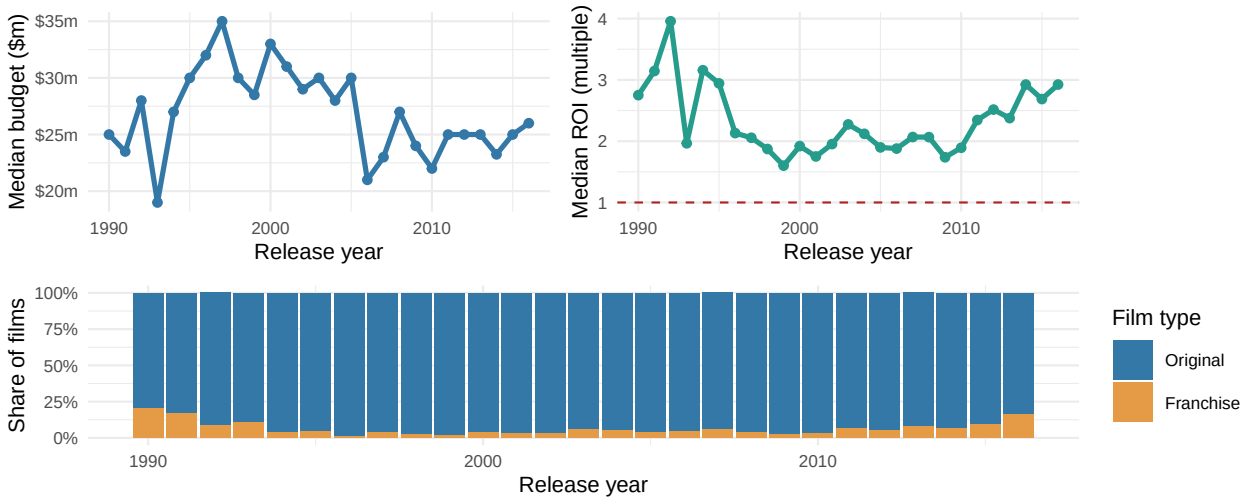


Figure 3: Median budgets, median ROI and film composition by release year, 1990–2016.

Production scale differs across genres: action and animation tend to need larger budgets, while horror and drama often operate at lower levels. ROI remains variable within every genre, so genre or budget alone should not be treated as a reliable predictor of financial success.

Finally, missing financial values are substantial: only **3,214 films** have both a valid budget and ROI. Comparisons should be viewed as exploratory, especially because box-office revenue is not the same as studio profit and ROI here does not deduct marketing or distribution costs.

2 Task 2: Franchise vs. Original: Investment Strategy

2.1 Introduction

Meridian Pictures must decide how to balance investment between Original films and Franchise films. This analysis compares their production budget distributions, estimates the difference in median budgets, and considers what the evidence implies for future investment decisions.

The analysis uses films released by the end of 2016. Films with missing release years and films released after 2016 were excluded. Budgets below $10,000 and nonpositive revenue values were treated as missing, consistent with the dataset documentation. The comparison uses all films with valid budget information and a known franchise classification, rather than restricting the analysis to the five focus genres.

Overall, Franchise films have higher median production budgets and higher median revenue per film than Original films. However, Original films account for a larger share of total reported revenue in this dataset. These findings suggest that Meridian should consider both budget allocation and financial performance when planning its future slate.

2.2 Budget Distribution Comparison

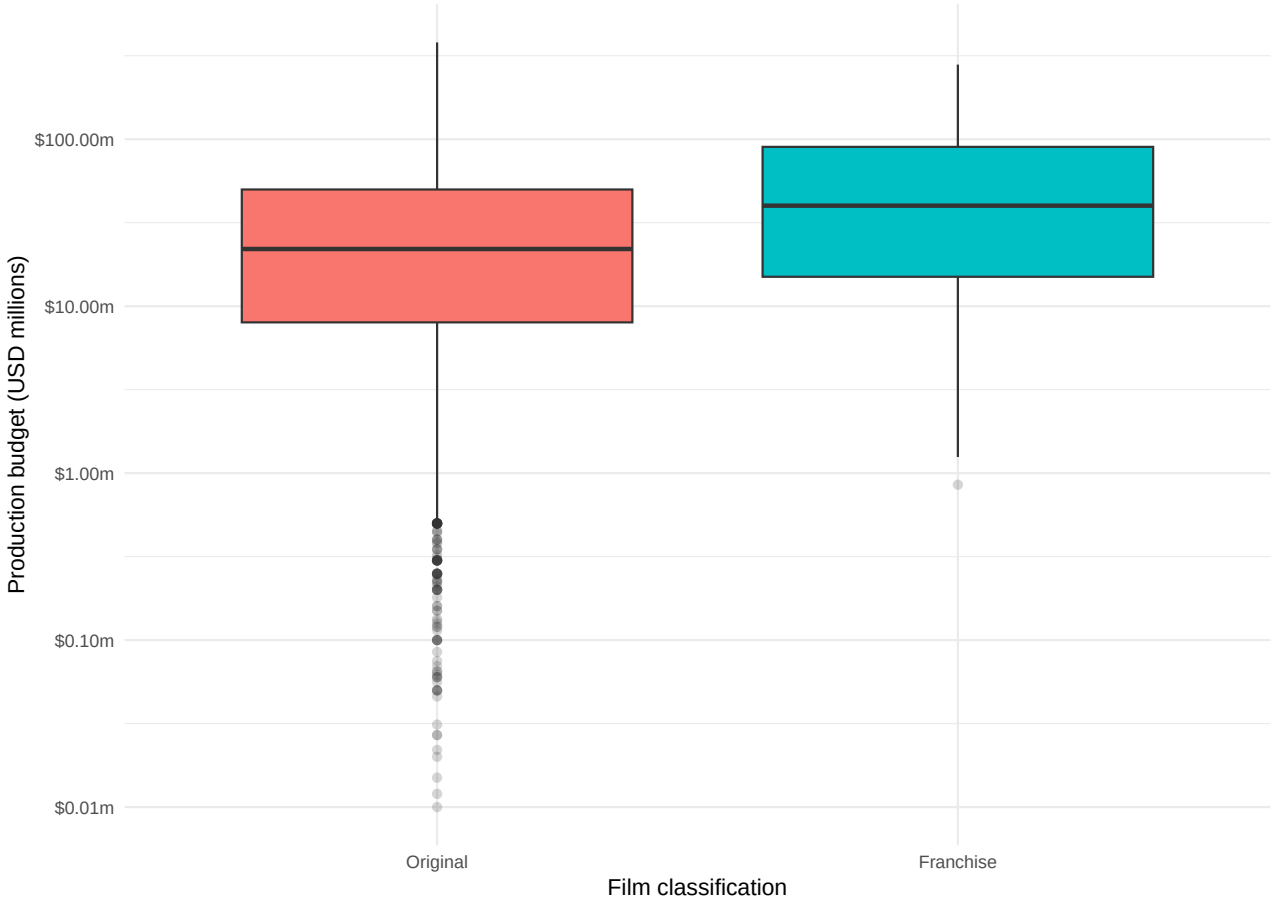


Figure 4: Production budget distributions for Original and Franchise films.

The boxplots show that Franchise films generally have higher production budgets than Original films, as reflected in their higher median. Numerous low-budget outliers appear in the Original group, suggesting that very small-budget projects are more common in this category. However, the distributions overlap, so some Original films have budgets comparable to those of Franchise films. These visual findings are examined further using a confidence interval for the difference in median budgets.

Table 4: Median production budget by film classification, in USD millions.

Film classification	Sample size	Median budget ($m)
Original	3498	22.00
Franchise	233	40.00

2.3 Interval Estimation

Table 5: 95% bootstrap interval for the difference in median production budgets (Franchise minus Original), in USD millions.

Lower ($m)	Upper ($m)
9.00	25.02

To estimate the uncertainty in the difference between median production budgets, we used a bootstrap percentile method with 1,000 resamples. In each resample, budgets were sampled with replacement separately for Original and Franchise films, preserving the sample size of each group. The difference between the resampled medians was calculated as Franchise minus Original, and the 2.5th and 97.5th percentiles of the bootstrap distribution were used to obtain a 95% interval estimate.

The resulting interval was approximately **$9.00 million to $26 million**, compared with an observed median budget difference of $18 million. As the interval lies entirely above zero, the results provide evidence that Franchise films have a higher median production budget than Original films (Table 5).

Table 6: Revenue by film classification, including total revenue, median revenue and revenue share.

Type	Films	Total revenue ($m)	Median revenue ($m)	Share (%)
Original	3,149	338,899.5	47.8	85.8
Franchise	227	56,198.4	152.2	14.2

2.4 Discussion

The findings partially support leadership's concern about the allocation of production budgets. Franchise films had a median budget of $40 million, compared with $22 million for Original films, and the 95% bootstrap confidence interval for the difference in medians (Franchise minus Original) was approximately $9.00 million to $25.03 million. As this interval lies entirely above zero, the results support a higher median budget for Franchise films. However, the budget distributions overlap, so not all Original films have smaller budgets than Franchise films.

The revenue results do not support the concern that Franchise films generate the majority of studio revenue in this dataset. Among films with reported revenue, Original films accounted for approximately 85.8% of total reported revenue ($338.9 billion), while Franchise films accounted for 14.2% ($56.2 billion). Nevertheless, Franchise films had a higher median revenue per film: approximately $152.2 million, compared with $47.8 million for Original films. This distinction suggests that the higher revenue typically earned by an individual Franchise film does not translate into a higher total revenue contribution from the Franchise group. These results should be interpreted in light of the unequal group sizes and the films with missing revenue data.

These revenue figures do not account for production or marketing costs and therefore cannot establish which group is more profitable.

2.5 Investment Strategy Recommendations

Meridian should gradually increase its investment in Original films while maintaining a core portfolio of established franchises. Rather than allocating budgets based solely on whether a project belongs to a franchise, the studio should assess each film according to its expected financial returns, production costs, commercial potential and contribution to creative diversity. Original projects could be developed

through a range of budget levels, allowing Meridian to explore new concepts without committing substantial resources to every untested idea.

Over the next decade, Meridian should evaluate the performance of both categories using profitability and return on investment, rather than revenue alone. It should also monitor the financial performance of new Original projects over time and adjust its investment allocations as evidence accumulates. This approach would allow the studio to pursue its creative-diversity goals while retaining established franchises as part of its investment portfolio. The available analysis does not establish an optimal budget allocation between the two categories, so Meridian should use ongoing performance reviews to guide future decisions rather than commit to a fixed proportion in advance.

3 Task 3: Modelling Budgets and Downside Risk

Meridian's Finance team needs two inputs for planning: a realistic probability model for production budgets, benchmarked against Universal Pictures, for its budget simulator; and a downside-risk figure for Action, its most capital-intensive genre. Downside risk is the 25th percentile of ROI (worldwide box-office revenue divided by production budget), the multiple below which the worst quarter of Action films fall.

3.1 (a) Universal Budget Distributions

We used the 243 Universal Pictures films (by `primary_company`) released from 1962 to 2016, excluding 17 with placeholder budgets under $10,000. Budgets (in $m) are strongly right-skewed: the middle half lie between $17m and $60.5m, but a few exceed $150m, pulling the mean ($44.5m) well above the median ($35m). Alongside the required normal distribution, we fitted gamma, log-normal and Weibull distributions, which take only positive values and allow for right skew, all by maximum likelihood using `fitdistr()` from the MASS package.

```
universal_films <- movies |>
  filter(release_year <= 2016, primary_company == "Universal Pictures", budget >= 10000) |>
  mutate(budget_m = budget / 1000000)

# Maximum-likelihood fits of the four candidate distributions (MASS::fitdistr)
universal_normal <- fitdistr(universal_films$budget_m, densfun = "normal")
universal_gamma <- fitdistr(universal_films$budget_m, densfun = "gamma")
universal_lognormal <- fitdistr(universal_films$budget_m, densfun = "lognormal")
universal_weibull <- fitdistr(universal_films$budget_m, densfun = "weibull")
```

3.1.1 Fit Diagnostics and AIC

Distribution	Parameters (units)	Estimate 1	Estimate 2	Log-likelihood	AIC
Gamma	shape, rate (per $m)	1.382	0.031	-1,158.1	2,320.1
Weibull	shape, scale ($m)	1.214	47.479	-1,158.3	2,320.6
Log-normal	meanlog, sdlog (log $m)	3.392	1.015	-1,172.6	2,349.2
Normal	mean, sd ($m)	44.481	37.432	-1,225.1	2,454.1

Table 7: Distributions fitted to Universal Pictures budgets, ordered by AIC (lowest first)

AIC rewards a close fit but penalises extra parameters, so lower is better. As all four distributions have two parameters, the ranking in Table 7 reflects fit alone. The gamma has the lowest AIC (2320.1), with the Weibull effectively tied (2320.6); the log-normal is 29 points worse and the normal is far behind (2454.1). AIC cannot separate the top two or show whether either fits well, so we also used Q-Q plots.

In Figure 5, the normal curves away at both ends: its lowest theoretical quantiles are negative (it gives about a 12% chance of a negative budget), and the largest budgets sit well above the line, so it understates expensive films. The log-normal errs the other way: its top theoretical quantile is over $500m, against Universal's actual maximum of $209m. The gamma and Weibull both track the line, but for the biggest budgets the Weibull points drift above it while the gamma stays close. As large budgets drive Finance's exposure, and the gamma also has the lowest AIC, we recommend the gamma distribution (shape 1.382, rate 0.031 per $m).

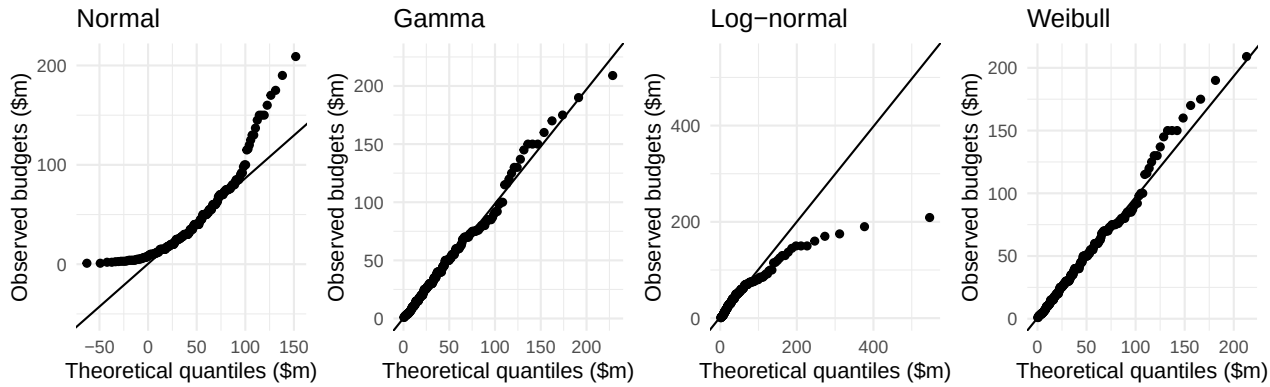


Figure 5: Q-Q plots of Universal Pictures budgets against each fitted distribution. Points close to the line indicate a good fit.

3.2 (b) Action Downside Risk and Intervals

For downside risk we used all studios: 588 of the 754 Action films (by primary genre, as in Task 1) released up to 2016, after excluding 97 with placeholder budgets and 69 with no reported revenue. The sample 25th percentile of ROI is 0.91 (Table 8), so a quarter of Action films returned less than 91 cents of box office for every dollar of budget.

```
action_films <- movies |>
  filter(release_year <= 2016, primary_genre == "Action", budget >= 10000, revenue > 0) |>
  mutate(roi = revenue / budget)
action_q25 <- quantile(action_films$roi, probs = 0.25)

# (1) Percentile bootstrap: resample the 588 films, take each resample's 25th percentile
set.seed(2420)
B <- 5000
boot_q25 <- tibble(simulation = 1:B, q25_roi = replicate(B, {
  boot_sample <- sample(action_films$roi, size = nrow(action_films), replace = TRUE)
  quantile(boot_sample, probs = 0.25)
}))
boot_q25_ci <- quantile(boot_q25$q25_roi, probs = c(0.025, 0.975))
```

```
# (2) Model-based: fit a log-normal, simulate 5,000 samples of 588 films, refit each
action_lognormal <- fitdistr(action_films$roi, densfun = "lognormal")
meanlog <- action_lognormal$estimate["meanlog"]
sdlog <- action_lognormal$estimate["sdlog"]
action_q25_model <- qlnorm(0.25, meanlog = meanlog, sdlog = sdlog)

set.seed(2420)
sim_q25 <- tibble(simulation = 1:B, q25_roi = replicate(B, {
  sim_fit <- fitdistr(rlnorm(nrow(action_films), meanlog = meanlog, sdlog = sdlog), densfun = "lognormal")
  qlnorm(0.25, meanlog = sim_fit$estimate["meanlog"], sdlog = sim_fit$estimate["sdlog"])
}))
sim_q25_ci <- quantile(sim_q25$q25_roi, probs = c(0.025, 0.975))
```

We estimated a 95% interval in two ways. The percentile bootstrap assumes only that the films are independent and representative of Action films. With seed 2420, we drew 5,000 resamples of the 588 ROI values with replacement, took each resample's 25th percentile, and used the 2.5th and 97.5th percentiles of these estimates as the interval. The model-based method assumes ROI is log-normal, a natural choice for a positive, right-skewed ratio. Fitting it by maximum likelihood gives meanlog = 0.482 and sdlog = 1.173 (the mean and standard deviation of log ROI), and a 25th percentile of 0.73. We simulated 5,000 samples of 588 films from this model (seed 2420), refitted the log-normal to each, and took the 2.5th and 97.5th percentiles of their 25th percentiles.

Method	25th percentile of ROI	95% lower	95% upper
Percentile bootstrap	0.91	0.80	1.05
Fitted log-normal model	0.73	0.66	0.82

Table 8: 95% interval estimates for the 25th percentile of ROI for Action films, from two methods

3.3 (c) Comparing the Estimators

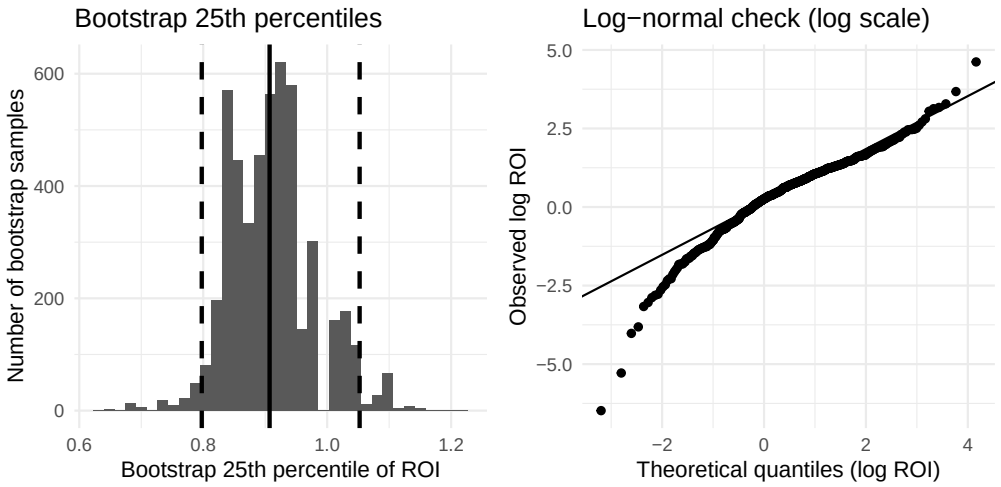


Figure 6: Left: bootstrap 25th percentiles of Action-film ROI (solid line: estimate; dashed lines: 95% interval). Right: Q-Q plot of log ROI against the fitted log-normal.

The two methods do not agree (Table 8): the bootstrap gives 0.91 (95% interval 0.80 to 1.05) and the model gives 0.73 (0.66 to 0.82). The intervals barely overlap, and the model's interval excludes the sample estimate. The most likely cause is the log-normal assumption. In the right panel of Figure 6, the lowest points fall well below the line, so the worst Action flops lost far more of their budget than a log-normal would predict. These extremes drag meanlog down (0.48, against a median log ROI of 0.68) and inflate sdlog, pulling the model's 25th percentile below what the films show; its narrower interval reflects trust in that shape, not extra information. A percentile is unaffected by how extreme the worst values are, so with 588 films and a bootstrap distribution centred on the estimate (left panel), we prefer the bootstrap interval. Two caveats apply: the bootstrap distribution is lumpy, as is usual for a percentile, and the self-reported data cut both ways. If some extreme flops have understated revenue, correcting them would push the true figure up, while if the 69 films without reported revenue performed poorly, including them would push it down.

3.4 (d) Finance Team Summary

Budget model. We recommend the simulator use a gamma distribution with shape 1.382 and rate 0.031 per $m. Of four shapes tested on 243 Universal films, using a model-comparison score (AIC) and visual checks, it fits best, including the few very expensive films, and it matches the average budget of about $44.5m. A normal (bell-shaped) curve would even generate negative budgets. As the budgets go back to 1962 without inflation adjustment, the model probably understates today's costs.

Action downside risk. Plan around an ROI of 0.91. Historically, one in four Action films earned less than 91 cents at the box office per dollar of budget; if future releases follow that pattern, roughly two of the eight contracted Action releases could fall below that level. The plausible range is 0.80 to 1.05, so use 0.80 for cautious stress tests. This figure comes from resampling the actual films; a fitted curve (0.73) did not match the worst results. ROI counts self-reported box office only, with no marketing costs.

4 Task 4: Building a Greenlight Risk Model

Using genre, franchise status, budget and crew size known before approval, we model the probability that a film fails to recover its production budget and assess three proposed films.

4.1 (a) Flop Definition and Model Choice

A *flop* has ROI (box-office revenue divided by production budget) below 1; missing ROI is unclassified. Of 4,801 films, 785 were flops, 2,429 non-flops and 1,587 lacked valid financial data. The observed flop rate among 3,214 classified films was 24.4%. We use logistic regression because it models a binary outcome and returns probabilities between zero and one, unlike unrestricted linear predictions; Poisson regression instead targets counts. Genre, franchise status, budget and crew size were available for 3,213 complete cases. Missing financial data may bias the sample.

4.2 (b) Frequentist and Bayesian Logistic Models

```
task4_frequentist <- glm(flop ~ primary_genre + franchise_status + budget_10m + crew_10,
                         data = task4_model_data, family = binomial)
```

Table 9: Selected frequentist logistic regression results

	Predictor	Odds ratio	95% CI lower	95% CI upper
franchise_statusFranchise	Franchise (vs original)	0.224	0.126	0.399
budget_10m	Budget (+$10 million)	0.960	0.936	0.985
crew_10	Crew (+10 members)	0.906	0.877	0.936

Interpretation

The frequentist logistic regression was fitted using 3,213 films with complete information. After accounting for genre, budget and crew size, franchise films had approximately 77.6% lower odds of flopping than original films (OR = 0.224, 95% CI: 0.126–0.399). A further $10 million in budget was associated with 4.0% lower odds of flopping (OR = 0.960, 95% CI: 0.936–0.985), while an additional 10 crew members was associated with 9.4% lower odds (OR = 0.906, 95% CI: 0.877–0.936). These findings suggest that franchise status, budget and crew size are associated with financial outcomes, although they do not establish causal relationships.

Limitations

Interpret these associations cautiously: the Foreign genre has only one classified film, causing complete separation and an unstable frequentist estimate. In addition, only 3,213 of 4,801 films were usable; non-random missing data could bias the results.

4.2.1 Bayesian Model and Diagnostics

Table 10: Selected Bayesian logistic regression estimates with 95% credible intervals.

Predictor	Median odds ratio (95% CrI)
Franchise (vs original)	0.236 (0.125–0.396)
Budget (+$10 million)	0.960 (0.935–0.983)
Crew (+10 members)	0.907 (0.878–0.936)

Bayesian findings and limitations

The Bayesian model used four Metropolis–Hastings chains (acceptance 35.5%–36.6%; minimum effective sample size 522.1; maximum upper R-hat 1.012), with broadly similar effects to the frequentist model. The sparsely observed Foreign genre remains highly uncertain despite prior regularisation (95% credible interval for its coefficient: −1.186 to 3.413).

4.2.2 Odds Ratios and Uncertainty Comparison

Table 11: Frequentist and Bayesian odds ratios with 95% uncertainty intervals.

Predictor	Frequentist OR (95% CI)	Bayesian OR (95% CrI)
Franchise	0.224 (0.126–0.399)	0.236 (0.125–0.396)
Budget (+$10m)	0.960 (0.936–0.985)	0.960 (0.935–0.983)
Crew (+10)	0.906 (0.877–0.936)	0.907 (0.878–0.936)

The models agree that franchise films have lower flop odds: frequentist OR 0.224 (95% CI 0.126–0.399) and Bayesian OR 0.238 (95% CrI 0.132–0.390). Budget and crew effects are likewise consistent (see Table 11). A confidence interval has a long-run coverage interpretation; a credible interval represents posterior uncertainty conditional on the model and priors. More than 99.9% of posterior draws imply lower odds for franchises, but these are associations, not causal effects.

4.3 (c) Classification Performance

Table 12: In-sample classification performance at two probability thresholds.

Threshold	Accuracy	Sensitivity	Specificity	Precision	Flops detected	Flops missed
50%	75.6%	0.6%	99.8%	50.0%	5	779
30%	66.6%	43.1%	74.2%	35.0%	338	446

Table 13: In-sample confusion matrix at the 30% threshold.

	Predicted non-flop	Predicted flop
Non-flop	1802	627
Flop	446	338

Classification performance

At the 50% classification threshold, accuracy was 75.6% but sensitivity just 0.6% (5 of 784 flops detected). Because flops comprise only about 24.4% of classified films, accuracy alone is misleading. Lowering the threshold to 30% increases sensitivity to 43.1% (338 flops), with 74.2% specificity, 35.0% precision and 66.6% accuracy (Table 12). The 627 false alarms make this a *review flag*, not an automatic rejection rule. These in-sample estimates may overstate performance on new films.

4.4 (d) Pitch Recommendations and Summary

```
task4_pitch_X <- model.matrix(
  ~ primary_genre + franchise_status + budget_z + crew_z,
  data = task4_pitches
)

task4_pitch_X <- task4_pitch_X[, colnames(task4_mh_draws), drop = FALSE]
```

Table 14: Bayesian predicted flop probabilities and 95% credible intervals for three hypothetical pitches.

Pitch	Flop probability	95% credible interval
Platform Zero	15.9%	11.1%–21.9%
The Meridian Protocol: Black Tide	4.7%	2.5%–8.0%
The Things We Never Recorded	28.8%	25.5%–32.6%

4.4.1 Recommendations for Three Pitches

The Meridian Protocol: Black Tide — Conditional greenlight

Lowest estimated risk: 4.7% (95% CrI 2.7%–7.9%). Conditional approval is reasonable, subject to checking the proposed $85 million budget and revenue forecasts.

Platform Zero — Conditional greenlight

Estimated flop risk is 16.0% (95% CrI 11.2%–22.2%). Consider conditional approval of this $9 million horror film after audience and distribution review.

The Things We Never Recorded — Hold for further review

Highest estimated risk: 28.8% (95% CrI 25.6%–32.6%). Hold the $18 million drama until its budget, audience and distribution assumptions are reassessed.

These recommendations are based on estimated flop risk rather than guaranteed outcomes. A lower predicted probability does not necessarily mean a project offers a better financial return.

4.4.2 Plain-Language Takeaway

For the Head of Production, the model is an early-warning tool, not a greenlight decision-maker. It favours *Black Tide*, then *Platform Zero*, while *The Things We Never Recorded* needs further review. Estimates omit marketing, audience demand and release strategy, measure flop probability rather than loss size, and have not been validated on new films. Use them alongside financial forecasts and professional judgement.

5 Appendix:

Our group's collaboration records, including task allocation, meeting notes, individual contributions, cross-review feedback and the final integration log, are all kept in this spreadsheet. The group contract and our generative AI declaration are also included in the spreadsheet.

Due to the page limit, we have placed supporting code for some analyses here. The most important code for each task is already shown within the task sections.

```
## Task 2: bootstrap interval for the difference in median budgets
set.seed(5242)

original_budgets <- movies_clean |>
  filter(franchise_status == "Original", !is.na(budget_m), budget_m > 0) |>
  pull(budget_m)
franchise_budgets <- movies_clean |>
  filter(franchise_status == "Franchise", !is.na(budget_m), budget_m > 0) |>
  pull(budget_m)

bootstrap_results <- tibble(replicate = 1:1000) |>
  mutate(
    boot_diff = map_dbl(replicate, function(i) {
      original_boot <- sample(x = original_budgets, size = length(original_budgets), replace = TRUE)
      franchise_boot <- sample(x = franchise_budgets, size = length(franchise_budgets), replace = TRUE)
      median(franchise_boot) - median(original_boot)
    })
  )

bootstrap_ci <- bootstrap_results |>
  summarise(
    lower = quantile(boot_diff, probs = 0.025),
    upper = quantile(boot_diff, probs = 0.975)
  )
```

```
# Task 2: revenue by film classification
revenue_summary <- movies_clean |>
  filter(!is.na(revenue), !is.na(franchise_status)) |>
  group_by(franchise_status) |>
  summarise(
    films_with_revenue = n(),
    total_revenue = sum(revenue),
    median_revenue = median(revenue),
    .groups = "drop"
  ) |>
  mutate(revenue_share = total_revenue / sum(total_revenue) * 100)
```

```
## Task 3: Q-Q plots of Universal Pictures budgets against each fitted distribution
qq_normal <- ggplot(universal_films, aes(sample = budget_m)) +
  stat_qq(distribution = qnorm, dparams = as.list(universal_normal$estimate)) +
  stat_qq_line(distribution = qnorm, dparams = as.list(universal_normal$estimate)) +
  labs(x = "Theoretical quantiles ($m)", y = "Observed budgets ($m)", title = "Normal") +
  theme_minimal()

qq_gamma <- ggplot(universal_films, aes(sample = budget_m)) +
  stat_qq(distribution = qgamma, dparams = as.list(universal_gamma$estimate)) +
  stat_qq_line(distribution = qgamma, dparams = as.list(universal_gamma$estimate)) +
  labs(x = "Theoretical quantiles ($m)", y = "Observed budgets ($m)", title = "Gamma") +
  theme_minimal()

qq_lognormal <- ggplot(universal_films, aes(sample = budget_m)) +
  stat_qq(distribution = qlnorm, dparams = as.list(universal_lognormal$estimate)) +
  stat_qq_line(distribution = qlnorm, dparams = as.list(universal_lognormal$estimate)) +
  labs(x = "Theoretical quantiles ($m)", y = "Observed budgets ($m)", title = "Log-normal") +
  theme_minimal()

qq_weibull <- ggplot(universal_films, aes(sample = budget_m)) +
  stat_qq(distribution = qweibull, dparams = as.list(universal_weibull$estimate)) +
  stat_qq_line(distribution = qweibull, dparams = as.list(universal_weibull$estimate)) +
  labs(x = "Theoretical quantiles ($m)", y = "Observed budgets ($m)", title = "Weibull") +
  theme_minimal()

qq_normal + qq_gamma + qq_lognormal + qq_weibull + plot_layout(nrow = 1)
```

```r
# Task 4: Bayesian model (priors and log-posterior)
task4_formula <- flop ~ primary_genre + franchise_status + budget_z + crew_z
task4_X <- model.matrix(task4_formula, data = task4_bayesian_data)
task4_y <- task4_bayesian_data$flop

task4_prior_sd <- rep(1.5, ncol(task4_X))
names(task4_prior_sd) <- colnames(task4_X)
task4_prior_sd[c("(Intercept)", "budget_z", "crew_z")] <- c(1, 0.5, 0.5)

task4_prior_mean <- rep(0, ncol(task4_X))
names(task4_prior_mean) <- colnames(task4_X)
task4_prior_mean["(Intercept)"] <- -1

task4_log_posterior <- function(beta) {
  eta <- drop(task4_X %*% beta)
  log_likelihood <- sum(task4_y * eta - (pmax(eta, 0) + log1p(exp(-abs(eta)))))
  log_prior <- sum(dnorm(beta, mean = task4_prior_mean, sd = task4_prior_sd, log = TRUE))
  log_likelihood + log_prior
}
```

```r
# Task 4: MCMC diagnostics and Bayesian odds ratios
task4_rhat <- coda::gelman.diag(task4_mh_chains, multivariate = FALSE)
task4_ess <- coda::effectiveSize(task4_mh_chains)

# Budget and crew were standardised, so divide by their SDs to get odds ratios per +$10m / +10 crew
bayesian_or_draws <- cbind(
  Franchise = exp(task4_mh_draws[, "franchise_statusFranchise"]),
  Budget = exp(task4_mh_draws[, "budget_z"] / task4_budget_sd),
  Crew = exp(task4_mh_draws[, "crew_z"] / task4_crew_sd)
)
apply(bayesian_or_draws, 2, quantile, probs = c(0.025, 0.5, 0.975))
```

```r
# Task 4: frequentist odds ratios and predicted flop probabilities for the three pitches
odds_ratios <- exp(coef(task4_frequentist))
wald_intervals <- confint.default(task4_frequentist)

task4_pitch_X <- model.matrix(~ primary_genre + franchise_status + budget_z + crew_z,
                              data = task4_pitches)
task4_pitch_X <- task4_pitch_X[, colnames(task4_mh_draws), drop = FALSE]
task4_pitch_prob_draws <- plogis(task4_mh_draws %*% t(task4_pitch_X))

task4_pitch_results <- data.frame(
  Pitch = task4_pitches$Pitch,
  Median_Probability = apply(task4_pitch_prob_draws, 2, median),
  Lower_95 = apply(task4_pitch_prob_draws, 2, quantile, probs = 0.025),
  Upper_95 = apply(task4_pitch_prob_draws, 2, quantile, probs = 0.975)
)
```